1N4007

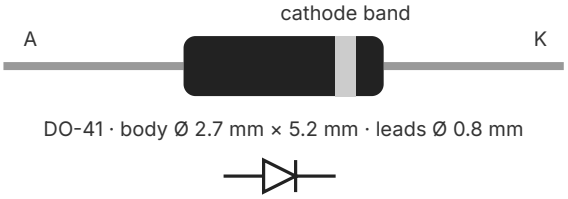
General-purpose silicon rectifier diode · 1 A · 1000 V

Datasheet summary compiled from typical manufacturer data (T_A = 25 °C unless noted) – for teaching purposes. Not an official manufacturer datasheet; values of real parts may differ slightly.

Features

- Diffused PN junction, glass-passivated
- High surge current capability
- Low reverse leakage current
- Typical use: rectifiers in power supplies, protection against reverse polarity

Package



DO-41 · body Ø 2.7 mm × 5.2 mm · leads Ø 0.8 mm

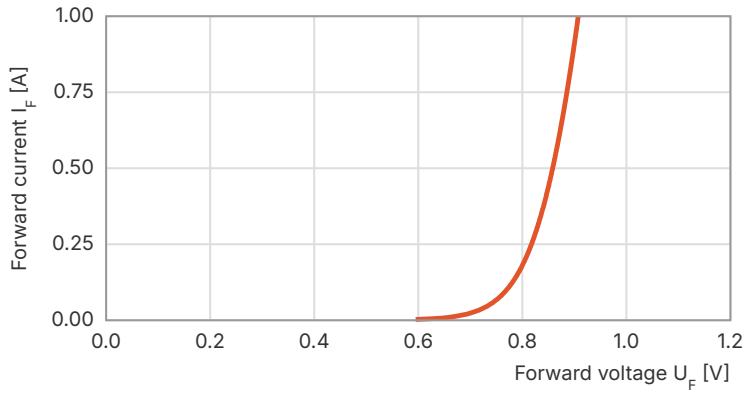
Absolute maximum ratings

Parameter	Symbol	Value	Unit
Repetitive peak reverse voltage	U_{RRM}	1000	V
RMS reverse voltage	$U_{R(RMS)}$	700	V
Average forward current (T_A = 75 °C)	$I_{F(AV)}$	1.0	A
Peak forward surge current, 8.3 ms half sine	I_{FSM}	30	A
Junction temperature	T_j	−65 … +175	°C

Electrical characteristics

Parameter	Symbol	Conditions	Min.	Typ.	Max.	Unit
Forward voltage	U_F	I_F = 0.5 A	–	0.86	–	V
Forward voltage	U_F	I_F = 1.0 A	–	0.91	1.1	V
Reverse current	I_R	$U_R = U_{RRM}$, T_j = 25 °C	–	–	5	µA
Reverse current	I_R	$U_R = U_{RRM}$, T_j = 100 °C	–	–	50	µA
Junction capacitance	C_j	U_R = 4 V, f = 1 MHz	–	15	–	pF

Typical forward characteristic (T_j = 25 °C)



Typical curve at 25 °C. The forward voltage rises with the current; at higher junction temperature it becomes slightly lower.